230701107 - Haritha.M

EXPERIMENT – 7

Code Injection Using Ptrace in Kali Linux (Simple Steps)

Aim:

To demonstrate code injection on a running application process using the ptrace system call in Kali Linux for understanding low-level process manipulation and security vulnerabilities.

Procedure:

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux virtual machine. Log in to the system.

Step 2: Create Target Program

Create a simple C program that runs continuously. This program will act as the target process for testing.

Step 3: Run Target Application

Execute the target program and identify its Process ID (PID), which is required to interact with the running process.

Step 4: Create Injector Program

Write a C program that uses the ptrace system call to attach to the target process and access its internal state such as registers or memory.

Step 5: Compile Programs

Compile both the target and injector programs using a C compiler so they can be executed.

Step 6: Attach to Target Process

Run the injector program and connect it to the target process using the PID. This allows control over the target process.

Step 7: Modify Execution

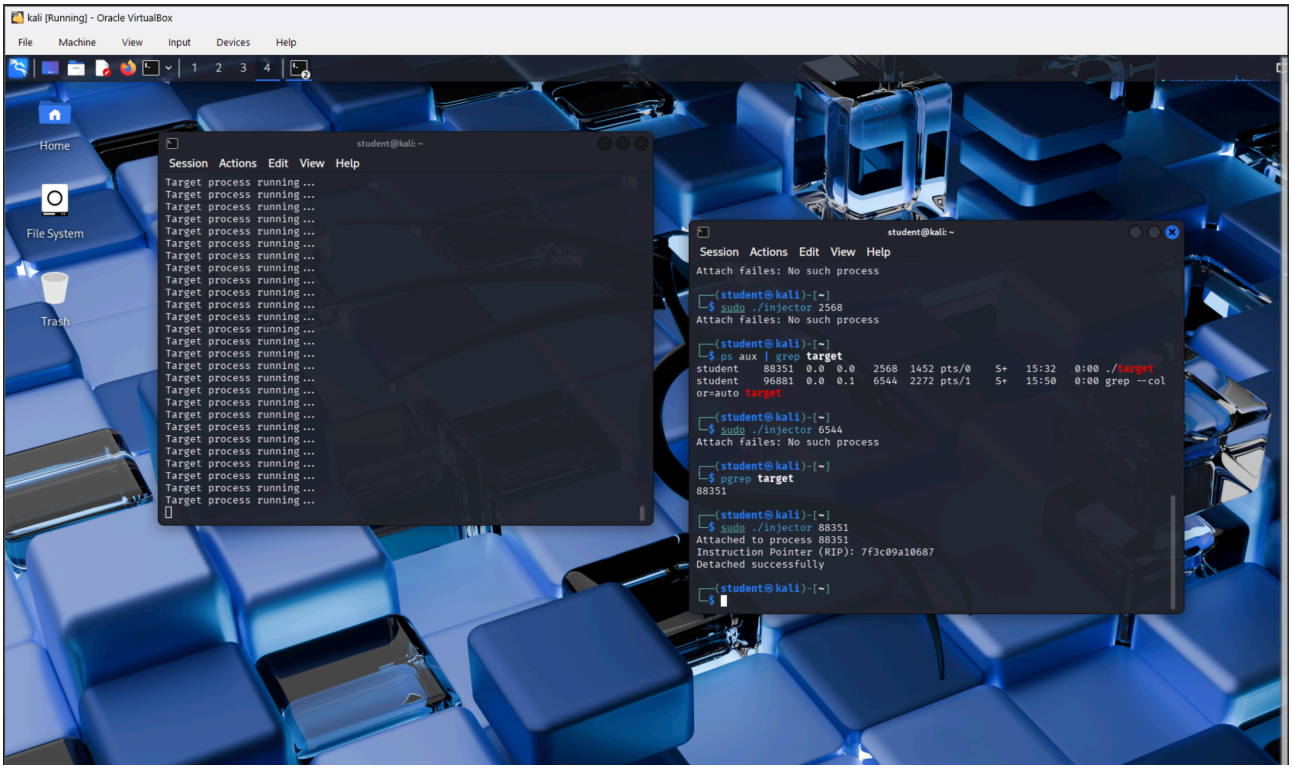
Change the target process behavior by altering its execution flow or memory content using ptrace.

Step 8: Observe Changes

Resume the process and observe how its behavior is modified due to the injection.

230701107 - Haritha.M

Output:



Result:

Code injection using ptrace was successfully demonstrated, showing how process execution can be controlled and highlighting the importance of system-level security and process isolation.